Paper 2

1 **(a) (i)** Use the kinematics equation $s = ut + \frac{1}{2}at^2$

$$h = \frac{1}{2}gt^2$$

$$g = \frac{2h}{t^2}$$

$$= \frac{2(2.66)}{0.740^2}$$

$$= 9.72 \text{ m s}^{-2}$$

(ii) 1. Percentage uncertainty of $h = \frac{\Delta h}{h} \times 100\% = \frac{1}{266} \times 100\%$

$$= 0.38\%$$

2. Percentage uncertainty of $t = \frac{\Delta t}{t} \times 100\% = \frac{0.005}{0.740} \times 100\%$

$$= 0.68\%$$

(b) $g = \frac{2h}{t^2}$

$$\frac{\Delta g}{g} = \frac{\Delta h}{h} + 2\left(\frac{\Delta t}{t}\right)$$

$$\Delta g = \left[\frac{0.38}{100} + 2\left(\frac{0.68}{100}\right)\right](9.72)$$

$$\Delta g = 0.169$$

$$= 0.2 \text{ m s}^{-2} \text{ (1 s.f.)}$$

$$g = (9.7 \pm 0.2) \text{ m s}^{-2}$$

(c) 1. The timer may have a zero error.

2. The timer may have been miscalibrated.

Note: Some given answers are not related to time. Students should not be explaining why g differ from “the correct” answer, or explaining the difference between precise and accurate.